

STEGANOGRAPHY LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
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Submission Platform:	Canvas
Report Date:	2026-02-27
Report Version:	1.0
Lab Title:	Steganography
Module Reference:	Steganography
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-02-27
Time Spent:	Approximately 1 hour
ENVIRONMENT	
Platforms and Systems:	
- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox. - Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Controller), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Domain Member), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation). - The work centered mainly on PLABWIN10 for 7-Zip + copy /b steganography and SilentEye, and on PLABKALI01 for Steghide and Stegosuite. - Remote access to the virtual machines was provided through the course platform.	
Context and Constraints:	
- The module focused on hiding data in images using 7-Zip copy /b technique, Steghide, Stegosuite, and SilentEye. - Practice Labs flagged the module as legacy content using deprecated software, so behavior could vary and some steps were best interpreted as guided sandbox demonstrations. - Windows Updates were disabled in the Practice Labs environment, so software behavior reflected a controlled legacy setup rather than a current hardened enterprise baseline. - The module aligned most closely with exam objectives 1.1 (Network and Communication Technologies), 3.1 (Information Security Controls), and 3.3 (Information Security Attack Prevention).	
Authorization and Scope:	
- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes. - No public IP space, real organizations, or unauthorized targets were involved. - No production infrastructure, third-party systems, or live services were contacted. - The actions documented here were performed for defensive understanding, steganography tool usage, and ethical hacking education within the course environment. - Restricted material, live credentials, and personal data are omitted or redacted from this report.	
Work Objectives:	
- Review steganography concepts and tools for hiding data in images. - Use 7-Zip + copy /b technique on PLABWIN10 to hide a zip file inside an image. - Use Steghide on PLABKALI01 to embed and extract a text file with password protection. - Use Stegosuite GUI on PLABKALI01 to embed and extract messages in images. - Use SilentEye on PLABWIN10 to embed messages in images.	
Steganography Concepts and Defensive Framing:	
- The lab identified three core techniques of steganography: file concatenation (7-Zip copy /b), command-line embedding (Steghide), and GUI embedding (Stegosuite and SilentEye). - Attack types reviewed included hiding text files and messages inside images without visibly altering them. - Tools demonstrated both Windows (7-Zip, SilentEye) and Linux (Steghide, Stegosuite) approaches, showing how steganography conceals data in plain sight. - Prevention concepts reviewed included regular file integrity checking, monitoring for unusual file size changes, and awareness training, reinforcing that steganography defense requires both technical controls and user vigilance.	

TECHNICAL ANALYSIS

7-Zip + copy /b Steganography on PLABWIN10:

- On PLABWIN10, File Explorer was used to create the C:\PLAB folder and two text files (PLAB1.txt and PLAB2.txt).
- Both text files were selected, right-clicked, and added to PLAB.zip using 7-Zip.
- An image file (PLAB.png) was copied from the Pictures folder into C:\PLAB.
- Command Prompt was opened in the C:\PLAB directory and the command `copy /b "PLAB.png"+PLAB.zip PLAB-new.png` was executed.
- The resulting PLAB-new.png file appeared identical to the original image but contained the hidden zip file.
- Opening PLAB-new.png with 7-Zip File Manager revealed the original PLAB1.txt and PLAB2.txt files inside.

Steghide on PLABKALI01:

- On PLABKALI01, `apt-get update` and `apt-get install steghide -y` were run to install Steghide.
- A plab directory was created with `mkdir plab` and `cd plab`.
- A secret.txt file was created with `touch secret.txt` and the image practice-labs.jpg was copied in.
- The command `steghide embed -cf practice-labs.jpg -ef secret.txt` was executed with password Passw0rd.
- The command `steghide extract -sf practice-labs.jpg` was then used with the same password to successfully extract the hidden secret.txt file.

Stegosuite and SilentEye:

- On PLABKALI01, `apt-get install stegosuite -y` was run and stegosuite gui launched.
- practice-labs.jpg was opened, the message "This is a confidential message." was entered with password Passw0rd, and Embed was clicked to create practice-labs_embed.jpg.
- The same image was reopened and Extract was used with the password to recover the hidden message.
- On PLABWIN10, SilentEye was installed and used to embed the message "This is a test!" into PLAB.png.

KEY COMMANDS, TOOLS, AND ARTIFACTS BY SYSTEM

PLABWIN10 — 7-Zip & SilentEye:

```
copy /b "PLAB.png"+PLAB.zip PLAB-new.png  
(7-Zip used to create PLAB.zip from hidden files)
```

```
SilentEye - Embed message "This is a test!" into PLAB.png
```

PLABKALI01 — Steghide & Stegosuite:

```
apt-get install steghide -y  
steghide embed -cf practice-labs.jpg -ef secret.txt  
steghide extract -sf practice-labs.jpg
```

```
apt-get install stegosuite -y  
stegosuite gui  
(embed "This is a confidential message." with password Passw0rd)
```

Primary Tools and Artifacts Observed:

- 7-Zip + copy /b created PLAB-new.png containing hidden PLAB.zip (extractable with 7-Zip).
- Steghide successfully embedded and extracted secret.txt inside practice-labs.jpg using password Passw0rd.
- Stegosuite GUI embedded and extracted the message "This is a confidential message." into practice-labs_embed.jpg.
- SilentEye embedded the message "This is a test!" into PLAB.png.
- The most important evidence of successful steganography was the visually identical images that contained hidden files/messages, verifiable only by the correct tool and password.

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered multiple steganography techniques: 7-Zip copy /b file concatenation, Steghide command-line embedding, Stegosuite GUI, and SilentEye.
- The recurring theme across the module was that steganography conceals data in plain sight without visibly altering the carrier file.

MITRE ATT&CK Mapping:

- Defense Evasion: T1027 Obfuscated Files or Information - data hidden inside images using Steghide/SilentEye.
- Exfiltration: T1048 Exfiltration Over Alternative Protocol - steganographic data concealment.
- Collection: T1119 Automated Collection - hiding documents for later extraction.

Detection Opportunities:

- Unusual file size increases, unexpected .aep or embedded image files, and unknown tools (Steghide, SilentEye) are strong indicators.
- File integrity monitoring and regular scanning for steganography tools provide early detection.
- Behavioral monitoring for copy /b commands or unusual image processing can flag activity.

Defensive Controls:

- Implement file integrity monitoring and regular hash verification to detect hidden data.
- Deploy endpoint detection/response (EDR) that flags steganography tools (Steghide, SilentEye).
- Use data loss prevention (DLP) solutions to scan images for embedded content.
- Train users to recognize suspicious image files and enforce least-privilege on systems.

Completion Verification:

- Work was completed across 7-Zip copy /b hiding, Steghide embed/extract, Stegosuite GUI, and SilentEye embedding.
- Completion was evidenced by successfully hidden/extracted files and messages in images.
- The actual commands and tool outputs used in the lab are preserved directly in this report for accurate documentation and reproducibility.

Key Findings:

- 1) Steganography allows data to be hidden in plain sight inside images without visible changes.
- 2) Simple techniques like 7-Zip copy /b are extremely effective and require no special software.
- 3) Steghide and Stegosuite provide password-protected embedding on Linux.
- 4) SilentEye offers an easy GUI method for Windows users to embed messages in images.
- 5) The strongest defensive takeaway was that regular file integrity checks and tool monitoring are critical to detecting steganography in an environment.

Issues / Deviations / Notes:

- The lab used deprecated software and a legacy environment, so exact tool behavior could vary.
- All activities were performed only inside the isolated Practice Labs sandbox.

Reflection:

- This lab tied together multiple steganography techniques (7-Zip copy /b, Steghide, Stegosuite, SilentEye) into a complete data-hiding workflow.
- The strongest takeaway was that steganography is dangerous because it conceals data in plain sight, making both technical detection and user awareness essential for defense.